An Enhanced Binary Cuckoo Search Algorithm Using Crossover Operators for Features Selection

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***Abstract*—One of the most important preprocessing steps is the determination of the most relevant subset of features in any dataset. This step is called "Feature Selection", which is considered an NP-Hard optimization problem. Cuckoo Search Algorithm (CSA) is a popular Nature-Inspired Meta-heuristic algorithm, which is used for handling continuous and discrete optimization problems. Although CSA has a great performance on many optimization problems, however, it lacks the balancing between the exploration and exploitation abilities. In this research, a binary cuckoo search algorithm with different types of crossover operators to improve is proposed. The crossover operators help the nest to discover different unexplored regions and avoid the issue of trapping in the local optima. All of the proposed versions have been applied on several datasets, and the results in terms of the classification accuracy and number of selected features were recorded. The experimental results demonstrated the effectiveness of the proposed algorithms in comparison to the original binary cuckoo search algorithm and different classifications.**

Keywords—Feature Selection, Binary Cuckoo Search Algorithm, Crossover Operators, Classification, Machine Learning.

I. INTRODUCTION

Nowadays is the era of huge data, where data is everywhere and in different fields from social media to education and health, The rapid expansion of the data dimensionality creates problems in handling it [1], [2]. In terms of classification, data classification represents a challenge in data mining and machine learning [3]. Sometimes the dataset containing redundant, irrelevant, and noisy data [3]. The redundant and irrelevant feature slows down the model and minimizes the classification accuracy, By selecting the relevant features, feature selection could solve this problem. Feature selection is used to improve the performance of the classifier and provide a more efficient and cost-effective classification [4]. Feature selection mission is challenging because of the huge search space. Different methods have been used for feature selection such as filter, wrapper, embedded method. In most circumstances, an exhaustive search is impossible in practice. To handle feature selection problems, a variety of search approaches have been used, including greedy search based on sequential forward selection (SFS) and sequential backward selection (SBS) [4]. This feature reduction strategy still has several flaws, such as local optima stagnation. Recently, Meta-heuristics in general, and Nature-Inspired algorithms have been implemented to handling different types of optimization problems in engineering and machine learning fields [5]. Evolutionary Computation (EC) methods are so well for their global searchability, and they have been utilized to treat feature selection problems [5]. Particle Swarm Optimization (PSO), Genetic Algorithms (GAs), and the Cuckoo Search Algorithm (CSA) are examples of (EC). CSA is easier to implement than GAs and PSO, has fewer parameters, less computationally expensive, and can converge faster [6]. Because of these benefits, the cuckoo search algorithm has been proposed as a potential solution to the problem. We provide a binary version of the Cuckoo Search Algorithm, The fundamental idea is to assign each feature a set of binary coordinates that indicates whether it will be included in the final set of features or not [7]. The main contribution of this paper is to design and develop a new binary version of the CS algorithm based on different crossover operators. The crossover operators help the nest escaping the local optima and enhance the searching performance of the algorithm. The following is a description of the paper's arrangement: Section 2 includes related work, Section 3 contains an overview of the original CS algorithm, and Section 4 contains a detailed explanation of the proposed algorithms. The Experiments are presented in section 5. The acquired results are presented in section 6. In section 7, you'll find the conclusion.

II. RELATED WORK

Al-Obaidi, Ahmed [8] (2013). Improved Scatter Search Using Cuckoo Search. This work proposes an improved scatter search technique. Scatter Search has greater diversity and intensity for potential answers, as well as random exploration of the problem's search space. The Traveling Salesman Problem was used to test the original and improved Scatter Search. The results show that the enhanced Scatter Search algorithms outperform the original Scatter Search

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algorithm in terms of performance. When compared to the original SS, the value of average fitness has increased by 23.2 percent. Alaa, Samer & Sadiq, Ahmed [9]. (2016). In this research, suggestion two enhancements to the cuckoo search algorithm for handling the Flexible Job-Shop Scheduling Problem (FJSP), one of which is based on Best Neighbors Generation (CS-BNG) and the other on Iterative Levy Flight (CS-ILF). To improve searching in the discrete state space, some adaptations to the CS algorithm's essential points have been made. The proposed algorithms have enhanced the accuracy and convergence rate of the solutions. For performance evaluation, the improved algorithms were tried on certain FJSP benchmark instances. The experimental results show that the enhanced algorithms are more effective than the basic cuckoo search algorithm. Abed-alguni, Bilal & Paul, David [10], (2018). For Numerical Optimization Problems, Using Different Mutation Operators to Hybridize the Cuckoo Search Algorithm. The main purpose of this study is to test the performance of the CS algorithm after replacing the Levy flight method with seven other mutation methods in the original CS algorithm. In terms of the accuracy and reliability of the acquired results throughout numerous simulations, the proposed CS variants were assessed using 14 common benchmark functions. The experimental results indicate that the CS with polynomial mutation produces more accurate and reliable outcomes than the other CS variants. Usman, Ali Muhammad et al.[1]. (2020), Feature selection aims to prevent incorrect features from being included while yet improving classification accuracy [14-16]. These objectives conflict with one another. Numerous studies have been done on feature selection problems, however, the majority of them are single-objective. Multi-objective optimization approaches are rapidly becoming the most effective methods for dealing with feature selection issues. They may simply find a balance between the features they want to use and the classification accuracy or error rate they want. Multi-objective feature selection using computational evolutionary approaches. The cuckoo optimization algorithm is one of the most widely used methods for solving feature selection problems.

III. OVERVIEW ON CUCKOO SEARCH ALGORITHM

Cuckoo Search (CS) is Nature Inspired Meta-heuristic Algorithm developed by Xin-She Yang and Suash Deb in 2009. Because of their beautiful sounds and aggressive reproductive technique, cuckoos are attractive birds. Several species of cuckoos, such as the Ani and Guira, lay their eggs in communal nests and may remove other people's eggs to enhance the chance of their eggs hatching. CS is based on a few cuckoo species' brood parasitism, in which they lay their eggs in the nests of other host birds. If a host bird recognizes the eggs as not being its own, it will either discard them or quit the nest and build a new one elsewhere each cuckoo egg indicates a new solution, and each egg in the nest represents a solution. If the new solution (cuckoo) is superior to the old, then it should be used instead of the current one in the nest. Rather than isotropic random walks, Levy flights improve this approach [11].

The following three principles were used to describe a typical cuckoo search.

1- Each cuckoo lays one egg and lays it in a nest that is chosen at random.

2- The best nests with the best eggs will be passed over to the next generation.

3- The number of available host nests is fixed, and the host bird discovers the cuckoo's egg with a probability of pa €. (0,1). The host bird has two options: get rid of the egg or give up the nest and build the nest again. To approximate this last assumption, a portion pa of the n host nests that are replaced with new nests might be used (with new random solutions). Cuckoo search is easy to use and provides a large search area instead of using a regular random walk, it employs levy flying for global search. The pseudo-code of the standard (CSA) [6] is shown in Algorithm (1).

Algorithm (1) Cuckoo Search Algorithm

INPUT: Red the dataset

OUTPUT: Best Selected Feature

Variable Definition: Maxitr: No. of iterations, Pop: Population Size, Pα: Fraction Probability, T: Current Iteration.

Begin

1. Generate an initial population of n host nest

2. Evaluate its fitness Fi.

3. Main while loop iteration of the algorithm.

While (T < Maxitr)

4. Get a cuckoo randomly using Levy Flight as Si

Choose a random nest as Sj

Evaluate the fitness of Si, Sj as Fi, Fj

If Fi > Fj Then

Replace Sj with Si

5. Abandon the nest with the worst fitness based on Pα, and generate a new nest

6. Rank the population and find the best nest as Best

End Loop

7. Return the Best

End

IV. ENHANCED CUCKOO SEARCH ALGORITHM

This section provides the details of the Enhanced Binary Cuckoo Search Algorithm (EBSCA) using different types of

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crossover operators where the classification error rate is used as a fitness function and the current solutions are evaluated according to the fitness function. Algorithm (2) shows the pseudo-code of the proposed (EBCSA) for feature selection.

Algorithm (2) Enhanced Cuckoo Search Algorithm

INPUT: Read the dataset

OUTPUT: Vector of feature selection, No. of

Variable Definition : Maxitr: No. of iterations, Pop: Population Size, Pα: Fraction Probability, Cp: Crossover Probability, T: Current Iteration. Fi, Fj: fitness function for solution**,** Best: best solution**,** O1, O2, O3, O4: offspring for Crossover Operator**,**

Begin

1. Initialize the Population randomly
2. Evaluate the fitness of each nest in the population
3. Main while loop iteration of the algorithm.

While (*T < Maxitr*)

4. Get a cuckoo randomly using Levy Flight as Si
   Choose a random nest as Sj
   Evaluate the fitness of Si, Sj as Fi, Fj
   If Fi > Fj Then
   Replace Sj with Si
5. Abandon the nest with the worst fitness based on Pα, and generate a new nest
6. Rank the population, find the best nest as Best
7. ***If*** the new Best is the same as the previous Then
8. O1, O2 = Uniform XO(Pop)
   O3, O4 = Two Point XO(Pop)
   Offspring = XOR(O1, O2, O3, O4)
9. Evaluate the fitness of Offspring as O Best
10. ***If*** O Best < Best then
    Best= O Best

End Loop

11 Return the Best

End

The enhanced algorithm can be explained in the following steps:

A. Initialization Population

Each nest represents a set of features, A binary string is used to encode the feature selection. For feature position, random values are created. If the value is 1, the feature is selected, otherwise, the feature is not [11].

B. Fitness Function

In terms of feature selection, the fitness function is crucial. The proposed method using classification error rate as a fitness function and the current solutions are evaluated according to the fitness function and the optimization goes. The used fitness function is outlined in "(1)" .

$$\text{Fitness (Xi)} = \text{N False / N} \quad (1)$$

Where *Xi* is the subset of features, *N False is the number of incorrectly classified data items* given *Xi* in the validation set and *N* is the total number of data in the validation set [12].

C. Finding New Solutions

For finding the new solution (xi^{t+1}), Cuckoo Search using levy flight as shown in "(2)" [6].

$$xi^{t+1} = xi^{t} + \alpha \oplus \text{Levy}(s, \lambda) \quad (2)$$

α is the step size, The symbol $\oplus$ represents an entry-wise multiplication and the step length follows levy distribution as shown in "(3)" .

$$\text{levy} \sim u = t^{-\lambda} \quad (3)$$

The variance is infinite, and the mean is infinite, the λ value falls between 1 and 3.

D. Crossover Operator

Each cuckoo lays one egg at a time in the original CS. The different types of crossover operators (Uniform, Two points) with (XOR) operators are used to create a new solution to increase exploration search space and convergence rate [9]. In the hybrid CS with crossover operators, every two-parent cuckoo will reproduce two children or cuckoo eggs to create different solutions. The final step is the use of the (XOR) operator to produce the final solution that is used in the next generation. The following operators are used in the enhanced binary cuckoo search algorithm

- Uniform Crossover Operator

Information exchange between parents based on probability values as shown in Fig. 1 ,

Parent1	1	0	1	0	0	0	1	1	1	0
Parent2	0	0	1	1	0	1	0	0	1	1
Mask	1	1	1	0	0	0	1	1	1	0
Offspring1	1	0	1	1	0	1	1	1	1	1
Offspring2	0	0	1	0	0	0	0	0	1	0

Fig 1. Uniform Crossover Operator

- Two Point Crossover Operator

Randomly two positions in the chromosomes are chosen, Everything between the two positions is swapped [13] as shown in Fig. 2 ,

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Fig 2. Two-point crossover operator

- XOR Operator

The XOR operator compares two bits and outputs a single bit. The concept is easy if all of the bits are equal, the outcome equals 0. The result is 1 if the bits are different.

V. EXPERIMENT DESIGN

Four datasets (Sonar, Robotics Execution Failure, Ionosphere, Breast Cancer Wisconsin) used in this study. The datasets were chosen to have different numbers of attributes, classes, and instances. which were downloaded from https://archive.ics.uci.edu/ml/datasets.php. In all datasets, instances are randomly divided into two groups: 70% of the training set and 30% of the test set. The classifiers were used (K-NN), (SVM), and (NB). The parameters of (EBCSA) are as follows: For each dataset, The number of execution times=10, with the number of starting initial cuckoos=30, the maximum number of iterations=100, and the fraction parameter P=0.25. Table I summarized each data set.

TABLE I. DATASET UTILIZED

Dataset	No. of Features	No. of Instance
Sonar	60	208
Robot Execution Failures	90	463
Ionosphere	34	351
Breast Cancer Wisconsin (Diagnostic)	32	569

VI. RESULTS ANALYSIS AND DISCUSSION

The outcomes of several methods are summarized in this section. First, standard classifiers (K-NN, NB, SVM), followed by Binary Cuckoo Search Algorithm with all classifiers, and last, an Enhanced Binary Cuckoo Search Algorithm (EBCSA) with all classifiers. Table II compares the results of the three approaches and shows that (EBCSA) with different classifiers (K-NN, NB, SVM) generated the best results in terms of accuracy. The values in the table represent the best value out of a total of ten implementations. In addition to the graph of the accuracy results using these classifiers are shown in Fig. 3, Fig. 4 and Fig. 5 .

TABLE II. EXPERIMENTAL RESULTS

	Dataset Accuracy			
Methods	**Robot Execute Failure**	**Breast Cancer**	**Ionosphere**	**Sonar**
K-NN	70%	75%	85%	73%
K-NN+BCSA	89%	95%	91%	93%
K-NN+EBCSA	**93%**	**96%**	**96%**	**95%**
SVM	63%	61%	92%	79%
SVM+BCSA	81%	98%	95%	90%
SVM+EBCSA	**85%**	**99%**	**96%**	**94%**
NB	98%	61%	92%	79%
NB+BCSA	100%	96%	95%	89%
NB+EBCSA	**100%**	**98%**	**96%**	**90%**

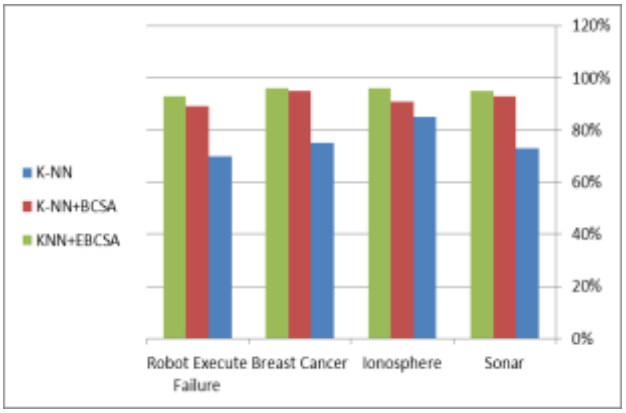


Fig 3. Accuracy Result of datasets with K-NN classifier

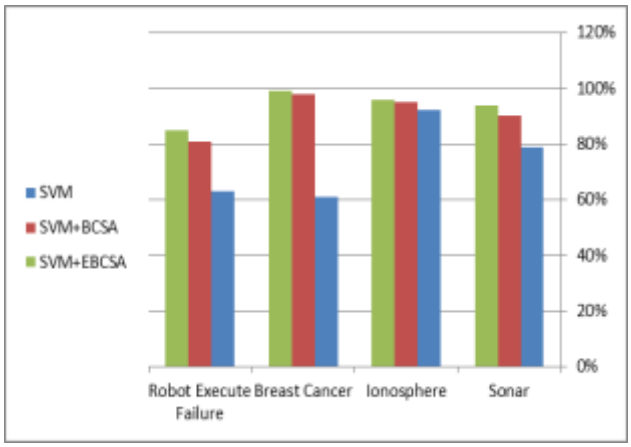


Fig 4. Accuracy Result of datasets with SVM classifier.

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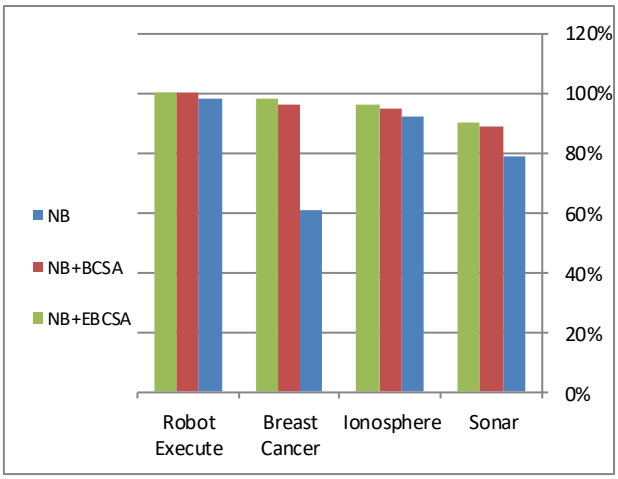


Fig 5. Accuracy Result of datasets with NB classifier

VII. CONCLUSIONS

This paper presented a Nature-Inspired algorithm Cuckoo Search (CS) and showed that the CS algorithm is more general and efficient than the Particle Swarm Optimization (PSO) and Genetic Algorithms (GAs). The algorithm was then optimized by using several types of crossover operators. To be used in different fields such as medical, engineering, and different applications. This paper aims to optimize CS performance so that it can more accurately and efficiently select relevant features from data. Experiments have shown that the suggested method produces the best results and has the highest accuracy, therefore it could be used instead of the standard cuckoo search algorithm in a variety of fields, particularly feature selection.

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